

STUDENT LEARNING MATERIAL

Probability Distributions

A complete guide to understanding how data is distributed — from the core concepts of mean and variance, through the major discrete and continuous distributions, to a real-world case study in game development.

WHAT'S INSIDE

Part 1 — Foundations: What a Distribution Is

Part 2 — Discrete Distributions: Uniform, Bernoulli, Binomial, Poisson

Part 3 — Continuous Distributions: Normal, Student's T, Chi-Squared, Exponential, Logistic

Part 4 — Case Study: Distributions in Practice (FIFA 19)

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Foundations: What a Distribution Is

Before studying specific distributions, we need the shared language: notation, mean, variance, standard deviation, and the difference between discrete and continuous data.

1.1 What Is a Probability Distribution?

DEFINITION

A **probability distribution** shows the possible values a variable can take and how frequently they occur.

Notation You Will Use Everywhere

- **Uppercase Y** — the actual outcome of an event (the random variable).
- **Lowercase y** — one specific possible outcome.
- **$P(Y = y)$** or simply **$P(y)$** — the probability that the outcome equals a particular value y . This is called the **probability function**.

EXAMPLE

Let Y be the number of red marbles drawn from a bag. Then the probability of drawing exactly five red marbles is written $P(Y = 5)$, or just $P(5)$.

How Probabilities Are Constructed

With a **finite** number of outcomes, we record the frequency of each unique value and divide by the total number of elements in the sample space.

With **infinitely many** outcomes, this is impossible — there are too many values to count.

Example: the time it takes for code to run. Any measurement (seconds, milliseconds, nanoseconds) can always be split in half for more precision, so no smallest unit exists. This is why continuous distributions need different tools, which we cover in Part 3.

1.2 The Two Defining Characteristics: Mean and Variance

Every distribution — finite or infinite — is defined using two characteristics:

Characteristic	Meaning	Population symbol	Sample symbol
Mean	The average value of the distribution	μ (mu)	$\bar{x}$ (x-bar)
Variance	How spread out the data is — measured by how far values sit from the mean	σ^2 (sigma-squared)	s^2

POPULATION VS SAMPLE

Population data = all the data. **Sample data** = just a part of it. The same data set can be both: surveying one whole department gives population data for the department, but only a sample of the whole company. Sample data uses different notation ($\bar{x}$ and s^2).

Standard Deviation — Making Variance Interpretable

Variance has one flaw: it is measured in **squared units** (seconds², kg², and so on), which usually has no direct interpretation. The fix is a third characteristic:

DEFINITION

Standard deviation is the positive square root of variance. Symbols: σ for a population, s for a sample. It is measured in the *same units as the mean*, so we can interpret it directly — which is why it is often preferred.

A key idea used throughout the course: any value between $\mu - \sigma$ and $\mu + \sigma$ falls **within one standard deviation of the mean**. The more congested the middle of the distribution, the more data falls inside that interval. The less data inside it, the more dispersed the data is.

$$\sigma^2 = E[(Y - \mu)^2] = E(Y^2) - \mu^2$$

The constant relationship between mean and variance, valid for any distribution. Specific distributions give us much more precise shortcut formulas.

1.3 Discrete vs Continuous — The Big Split

	Discrete	Continuous
Outcomes	Finitely many distinct outcomes	Infinitely many consecutive outcomes
Examples	Rolling a die, picking a card	Time and distance in track and field
Represented by	Table, graph, or formula	Graph of a curve (PDF) — a table is impossible
Graph looks like	Separate, unconnected bars	A smooth curve

Defining a Distribution — Standard Notation

We write the **variable name**, then the **tilde sign** ($\sim$), then a **capital letter for the distribution type**, and finally the **key characteristics in parentheses** (usually mean and variance, but this varies by type).

EXAMPLE

$X \sim N(\mu, \sigma^2)$ reads as: "Variable X follows a normal distribution with mean μ and variance σ^2 ."

Working With Discrete Distributions

- Every unique outcome must have a probability assigned to it (think of a dartboard — each region's probability depends on how big its interval is).
- To find the probability of an **interval**, simply **add up** the probabilities of all values inside it.
Example: $P(\text{three spades or fewer}) = P(0) + P(1) + P(2) + P(3)$.
- A useful peculiarity of discrete events: $P(Y \leq y) = P(Y < y + 1)$. "Three spades or fewer" is exactly the same as "fewer than four spades."

Discrete Distributions

Four key distributions for events with countable outcomes: Uniform, Bernoulli, Binomial, and Poisson.

2.1 Uniform Distribution — All Outcomes Equally Likely

NOTATION

$X \sim U(a, b)$ — for example, $X \sim U(3, 7)$ reads "X follows a discrete uniform distribution ranging from 3 to 7."

When it applies: events where all outcomes have **equal probability** (also called *equiprobable* outcomes).

EXAMPLES

Rolling a standard six-sided die — every value from 1 to 6 has probability $1/6$, so the graph is six equally tall bars. Also: many gambling events, or choosing between three identical chocolate bars a friend offers you.

KEY LIMITATION

Because all outcomes are equally likely, the mean and variance are **uninterpretable and have no predictive power**. For a die, the mean is 3.5 and the variance is $105/36$ — but neither value tells you anything useful about what will happen next.

2.2 Bernoulli Distribution — One Trial, Two Outcomes

NOTATION

$X \sim \text{Bern}(p)$ — "X follows a Bernoulli distribution with probability of success equal to p ."

When it applies: any event with **one trial and exactly two possible outcomes** — a coin flip, a single true/false quiz question, a two-party vote. Any two-outcome event can be transformed into a Bernoulli event by assigning one outcome to be "true" and the other "false," even if the outcomes are not equally likely.

Conventions

- The graph is just **two bars**: one rising to p , the other to $1 - p$.
- We usually denote the **higher** probability as p , and assign the value **1** to that favored outcome. That way the expected value expresses the likelihood of the favored event.

$$E(X) = p \quad \sigma^2 = p(1 - p)$$

The variance is always $p(1 - p)$ — regardless of the expected value. This is the first example of the elegant statistics distributions can possess.

WORKED EXAMPLE — THE UNFAIR COIN

A weighted coin lands on tails 60% of the time. Assign tails = 1, so $p = 0.6$.

Expected value: $E(X) = 0.6$. Variance: $0.6 \times 0.4 = \mathbf{0.24}$.

2.3 Binomial Distribution — Repeating the Bernoulli Trial

NOTATION

$X \sim B(n, p)$ — for example, $B(10, 0.6)$ reads "X follows a binomial distribution with 10 trials and a 0.6 likelihood of success on each individual trial." A Bernoulli distribution is simply a binomial with a single trial: $\text{Bern}(p) = B(1, p)$.

The core idea: binomial events are a **sequence of identical Bernoulli events**.

INTUITION

A surprise pop quiz has 10 true/false questions you haven't prepared for. Guessing **one** question is a Bernoulli event. Guessing the **entire quiz** is a binomial event. The Bernoulli expected value tells you which outcome to expect on a single trial; the binomial expected value tells you *how many times* to expect that outcome over all trials.

The graph: for n trials, there are **$n + 1$ bars** — one for each possible count from 0 to n . (Two flips of a coin → three bars: 0, 1, or 2 tails.)

The Probability Function

To get the favored outcome exactly y times out of n trials, three things must be accounted for: success y times (p^y), the alternative outcome the remaining $n - y$ times ($(1 - p)^{n-y}$), and **all the different orders** in which those successes can occur — which is exactly the combinations formula from combinatorics. (There are 3 different ways to get tails exactly twice in three coin flips.)

$$P(y) = C(n, y) \cdot p^y \cdot (1 - p)^{n-y}$$

$C(n, y)$ = the number of combinations of picking y elements out of n . This is why combinatorics is a fundamental part of probability.

WORKED EXAMPLE — THE STOCK PRICE

You own one General Motors stock. Historically, there is a 60% chance the price goes up on any given day and 40% it drops. What is the probability of exactly **3 increases in a 5-day work week**?

Plug in $y = 3$, $n = 5$, $p = 0.6$:

$$P(3) = C(5, 3) \times 0.6^3 \times 0.4^2 = 10 \times 0.216 \times 0.16 = \mathbf{0.3456}$$

There is a **34.56%** chance of exactly three increases during the week.

$$E(X) = n \cdot p \quad \sigma^2 = n \cdot p \cdot (1 - p)$$

Stock example: variance = $5 \times 0.6 \times 0.4 = 1.2$, so standard deviation ≈ 1.1 . Knowing $E(X)$ and σ lets us make more accurate forecasts.

2.4 Poisson Distribution — How Often in an Interval?

NOTATION

$Y \sim Po(\lambda)$ — "Y follows a Poisson distribution with lambda equal to 4." Lambda (λ) is the single parameter.

When it applies: when we know **how often an event occurs on average** over a specific interval of time or distance, and we want the likelihood of a particular frequency. It answers: "how unusual is this frequency for this interval?"

EXAMPLES

A firefly lights up 3 times per 10 seconds on average — how likely is it to light up 8 times in 20 seconds? Or: a basketball player averages 35 points per game — how likely is he to score exactly 12 points in the first quarter?

The graph: starts from zero (an event can't happen a negative number of times) but has **no upper cap** on how many times it could occur.

$$P(y) = (\lambda^y \cdot e^{-\lambda}) / y!$$

e is Euler's number ≈ 2.72 . Remember: $e^{-\lambda}$ is just $1 / e^\lambda$.

WORKED EXAMPLE — STUDENT QUESTIONS

Your online course usually gets about **4 questions per day** ($\lambda = 4$). Yesterday students asked **7**. How likely was that exact number?

$$P(7) = 4^7 \times e^{-4} / 7! = 16,384 \times 0.0183 / 5,040 \approx \mathbf{0.06}$$

Only about a **6%** chance of receiving exactly seven questions — so yesterday really was an unusual spike.

$$E(Y) = \lambda \quad \sigma^2 = \lambda$$

Both the mean and the variance equal lambda — another example of elegant statistics we can take advantage of.

For interval probabilities, the approach is the same as any discrete distribution: sum the probabilities of the individual values inside the interval.

Continuous Distributions

When outcomes are infinite, we switch from bars to curves — and from sums to areas. PDF, CDF, and the five key continuous distributions.

3.1 PDF and CDF — The New Toolkit

Probability Density Function (PDF)

DEFINITION

The **PDF**, written $f(y)$, depicts the associated probability density for every possible value y . Its value is always ≥ 0 . Its graph is the **probability distribution curve** — imagine the bars of a discrete graph getting narrower and narrower until a smooth curve traces along their tops.

Why Individual Values Have Probability Zero

With an infinite sample space, the denominator of the "favored over all" formula becomes enormous — so the probability of any single exact value is essentially **zero**. Consequence:

KEY FACT

For continuous variables, $P(Y > x) = P(Y \geq x)$. Example: the probability of running a mile in *under* six minutes is the same as running it in *at most* six minutes — because finishing in *exactly* 6:00.000... has probability zero.

Cumulative Distribution Function (CDF)

DEFINITION

The **CDF**, written $F(y)$ (capital F), represents the probability of the variable being **less than or equal to** a specific value: $F(y) = P(Y \leq y)$. Boundary values: $F(-\infty) = 0$ and $F(+\infty) = 1$.

Discrete distributions have CDFs too, but they're used less — with discrete data we can always just add up the individual PDF values instead.

How PDF and CDF Connect

Direction	Operation
PDF $\rightarrow$ CDF	Integrate the PDF from $-\infty$ to y
CDF $\rightarrow$ PDF	Take the first derivative of the CDF with respect to y
Probability of interval [A, B]	Area under the PDF curve from A to B (an integral — free tools like WolframAlpha can compute these)

Expected Value and Variance for Continuous Variables

Since each individual value has probability zero, summation no longer works. Instead:

- **Expected value** = the integral of $y \cdot f(y)$ over the interval from $-\infty$ to $+\infty$.
- **Variance** = the same short formula as before: $\sigma^2 = E(Y^2) - [E(Y)]^2$.

3.2 Normal Distribution — Nature's Favorite

NOTATION

$X \sim N(\mu, \sigma^2)$ — "X follows a normal distribution with mean μ and variance σ^2 ."

When it applies: the normal distribution frequently appears in **nature and everyday life** — hence the name. Example: the weight of adult male lions averages around 150–250 kg. Specimens outside that range exist, but they are the exception — **outliers** — and the more data we gather, the smaller a share of it they represent.

Key Characteristics

- **Bell-shaped graph** — most data is centered around the mean; values further away are less likely.
- **Symmetric** about the mean — values equally far away in opposite directions are equally likely. (If the mean is 400 lbs, a 350 lb lion and a 450 lb lion are equally likely — both 50 lbs from the mean.)
- $E(X) = \mu$. The variance σ^2 is usually given; if not, derive it from $\sigma^2 = E(X^2) - [E(X)]^2$.

The 68 - 95 - 99.7 Law

For any normally distributed event: **68%** of outcomes fall within 1 standard deviation of the mean, **95%** within 2, and **99.7%** within 3. Outliers are extremely rare — and just knowing data is normal already tells us a great deal about it.

3.3 Standardizing the Normal Distribution

First: What Is a Transformation?

DEFINITION

A **transformation** alters every element of a distribution to get a new distribution with similar characteristics. For normal distributions, addition, subtraction, multiplication, and division all keep the distribution normal.

Operation	Effect on graph
Add a constant (e.g., +3)	Shifts the graph right
Subtract a constant	Shifts the graph left
Multiply by a constant > 1	Graph shrinks (tighter)
Divide by a constant > 1	Graph expands (wider)
Multiply/divide by a value between 0 and 1	Opposite effect — dividing by $\frac{1}{2}$ is the same as multiplying by 2, so the graph shrinks

Standardizing

DEFINITION

Standardizing is the special transformation that makes the expected value equal **0** and the variance equal **1**. The result is the **standard normal distribution**, denoted Z .

How to do it: first subtract the mean μ from every element (shifting the graph so its mean is 0), then divide every element by the standard deviation σ (making the SD equal 1):

$$Z = (Y - \mu) / \sigma$$

Every Z value = how many standard deviations the original value sits from the mean. If Y is 2.3 standard deviations away from the mean, its Z equals 2.3.

The most common CDF values of the standard normal are summarized in the **standard normal distribution table**, also known as the **Z-score table**. This is what makes standardizing so convenient — a single transformation lets you analyze *any* normal distribution using one table.

WHEN NORMAL FAILS

The normal distribution **requires a lot of data**. With a limited sample, outliers can drastically distort the analysis. Rule of thumb: with **fewer than 30 entries**, avoid assuming normality — use the Student's T distribution instead.

3.4 Student's T Distribution — Normal's Small-Sample Cousin

NOTATION

$Y \sim t(k)$ — lowercase t, followed by a single parameter k called the **degrees of freedom**.
Example: $t(3)$.

When it applies: situations where we would assume a normal distribution if not for the limited number of observations. Example: average lap times for an entire Formula 1 season follow a normal distribution — but the lap times for just the first lap of the Monaco Grand Prix follow a Student's T.

Key Characteristics

- Bell-shaped and symmetric, like the normal — but with **fatter tails**. In a small sample, any extreme value represents a much bigger share of the total, so the distribution must accommodate values far from the mean.
- Requires **degrees of freedom** on top of the mean and variance.
- **Expected value** = μ (as long as there are at least 2 degrees of freedom).
- **Variance** = $s^2 \times k / (k - 2)$.
- Frequently used for **hypothesis testing with limited data**, and has its own table of important CDF values.

3.5 Chi-Squared Distribution — The Statistician's Tool

NOTATION

$Y \sim \chi^2(k)$ — capital Greek chi squared, with degrees of freedom k. Example: $\chi^2(3)$.

When it applies: very few real-life events follow it directly. Chi-squared appears mostly in **statistical analysis** — hypothesis testing, confidence intervals, and especially determining the **goodness of fit** of categorical values.

Key Characteristics

- The first **asymmetric** continuous distribution we meet — it consists only of **non-negative** values, so its graph starts from zero on the left.
- Typically **highly skewed to the right**.
- Like the normal and Student's T, it comes with a **table of known values**.
- **Expected value** = k. **Variance** = 2k.

ACCURACY NOTE

The lecture mentions squaring a Student's T to get a chi-squared. The precise statement in standard statistics: squaring a **standard normal** variable gives a chi-squared with **1 degree of freedom**. Keep this precise version in mind for exams and further courses.

3.6 Exponential Distribution — Fast Early, Then Fading

NOTATION

$X \sim \text{Exp}(\lambda)$ — followed by a rate (scale) parameter λ . Example: $\text{Exp}(\frac{1}{2})$.

When it applies: variables whose probability **starts high, decreases rapidly early on, then plateaus**.

EXAMPLE — YOUTUBE VIEWS

A new vlog gets most of its views in the first day or two after release. The total keeps growing, but the number of *new* daily views shrinks as the video becomes outdated or the creator posts new content. A random viewing is far more likely to have occurred close to release than in any later period.

Key Characteristics

- **PDF:** starts very high, drops sharply — the curve resembles a boomerang with each handle lining up along the axes.
- **CDF:** also boomerang-like but rotated — climbs and plateaus approaching 1. The rate parameter λ controls how fast the plateau is reached and how spread out the graph is.
- **Expected value** = $1/\lambda$. **Variance** = $1/\lambda^2$.
- Unlike the normal or chi-squared, there is **no table of known values** — so in practice, we often **transform** the data instead.

THE LOG TRANSFORM

A very common move in data analysis: take the **natural logarithm** of every element of exponentially-shaped data. The transformed data becomes much closer to normally distributed, which makes it suitable for running **linear regressions**. This is one of the most frequently performed transformations in statistics.

3.7 Logistic Distribution — Forecasting a Cutoff

NOTATION

$Y \sim \text{Logistic}(\mu, s)$ — the whole word "Logistic," followed by two parameters: the **mean** (also called the **location**) and a **scale** parameter. Example: $\text{Logistic}(6, 3)$.

When it applies: determining how **continuous inputs affect the probability of a binary outcome** — commonly in forecasting competitive sports where there are only two outcomes, victory or defeat.

EXAMPLE — THE TENNIS SERVE

Does serve speed decide the point? Faster serves give opponents less time to react — but reaching top speed sacrifices accuracy. So the relationship is **not linear**: there is an **optimal speed** where the serve is fast enough yet still accurate. Most converted points

cluster around that speed.

In the distribution: the **mean** represents the optimal speed, and the **scale** represents how forgiving the player's range is. A great server like Serena Williams can hit excellent serves further from the optimal speed — so her PDF is more spread out than her opponents'.

Key Characteristics

- **PDF:** looks similar to the normal distribution — the mean dictates the center, the scale dictates the spread.
- **CDF:** an S-shaped curve — starts slow, picks up quickly near the mean (where the probability of success rises drastically), then plateaus near 1. The **smaller** the scale, the **later** the curve picks up but the **quicker** it reaches values close to 1.
- **Variance** = $s^2 \times \pi^2 / 3$.

Case Study: Distributions in Practice

You are the head project manager at EA Games supervising FIFA 19. Watch five different distributions show up in one real project — and drive real business decisions.

4.1 Is the Game Balanced? — Normal Distribution

The question: a competitive game must be balanced — no team or player should always be the preferred option, so good and poor players should be equally represented.

The analysis: the *overall* column rates each player 1–100 (a weighted average of individual stats, where the weights depend on position — acceleration matters for wingers, tackling for center backs). Building a histogram in Excel (Insert → Insert Statistic Chart → Histogram, bin width set to 1) shows a **bell-shaped graph resembling a normal distribution**.

- **"But aren't the values discrete?"** Continuous variables can take discrete values (but not vice versa) — and since overall ratings are rounded averages, they are approximations rather than truly discrete values.
- **Thin tails** = few outliers. Realistic: very few professionals are exceptionally good or bad at *everything*, and even the weakest professional far exceeds the average person — which is why values start around 50, not 0.
- **Symmetry** confirms the ratings are symmetrically distributed.

The decision: the game is balanced and acceptable for competitive play.

4.2 Testing a Small Sample — Student's T

Taking just the **first 30 players** by ID and building a histogram (bin size adjusted to 3, otherwise every value appears only once or twice and the histogram is useless) gives a shape that slightly resembles a normal — but with **fatter tails**. With limited observations, this sample follows a **Student's T distribution**. Since it is still symmetric, even the small sample confirms the balance goal.

4.3 Two Peaks in One Stat — Reading a Bimodal Graph

The *shot power* histogram shows **two peaks** — like two graphs side by side, one group with a mean around 21 and another around 65. The reason: **goalkeepers**. Their essential stats differ completely from outfield players, so they score distinctly lower on non-goalkeeper stats. A goalkeeper stat like *GK diving* shows the two clusters even more clearly. Examine the goalies alone, and their values are normally distributed again — the game is indeed balanced.

LESSON

When a histogram shows two peaks, suspect **two distinct groups mixed in one data set**. Split them, and each group often follows a familiar distribution.

4.4 Realistic Player Ages — Poisson

In real soccer, young professionals outnumber veterans (injuries, forced retirement, quitting after too much bench time, choosing university instead). The *age* histogram should reflect that. Age starts at a minimum of **16** — the youngest age for a professional first-team contract — which can be viewed as the **origin of a Poisson distribution**. Each bar shows the likelihood of a player being a specific age, and since Poisson is skewed, younger players outnumber older ones — adding a layer of realism to the game.

4.5 Competitor's Video Views — Exponential

The setup: a competitor posts free preview videos every Monday. An intern tracked their daily views for a month.

The analysis: a scatter plot of days vs daily views starts very high (~100,000 views/day) and drops fast — to ~20,000 within a week, ~10,000 once the next video arrives, then a few thousand by day 14. This resembles an **exponential distribution**. Adding an **exponential trend line** in Excel (not the default linear one) fits the data points accurately — this trend line represents the PDF. Plotting **cumulative total views** shows a curve rising at a decreasing rate and plateauing — matching the expected CDF shape.

The decision: each video loses relevancy quickly, so the campaign only works short-term. Advise marketing to release similar videos **only during the last month before launch** — generating maximum attention without exhausting customer interest.

4.6 Who Buys Premium? — Logistic

The setup: a survey asks customers whether they hold a premium fan-club membership. The data: customer age plus a 1 (member) or 0 (not a member).

The analysis: a scatter plot shows most people **under 34** without a membership and most **over 34** with one (with normal real-world exceptions). The likelihood of membership rises sharply near a specific value — the signature of a **logistic distribution** with **location ≈ 34** , plausibly the age at which customers reach financial stability.

The decision: target customers above 34, who are more likely to spend — for example, by releasing more expensive legend player cards featuring stars who retired in the past 20 years.

Quick Reference & Review

Everything on one table, plus review questions to test yourself.

5.1 Distribution Cheat Sheet

Distribution	Notation	Use it when...	E(X)	Variance
Uniform (discrete)	$U(a, b)$	All outcomes are equally likely (die roll)	Uninterpretable	Uninterpretable
Bernoulli	$Bern(p)$	One trial, two outcomes (coin flip)	p	$p(1 - p)$
Binomial	$B(n, p)$	n identical Bernoulli trials (whole quiz)	$n \cdot p$	$n \cdot p \cdot (1 - p)$
Poisson	$Po(\lambda)$	Frequency of an event in a fixed interval	λ	λ
Normal	$N(\mu, \sigma^2)$	Natural phenomena, large samples, bell shape	μ	σ^2 (given)
Student's T	$t(k)$	Would be normal, but sample < 30 (fat tails)	μ ($k \geq 2$)	$s^2 \cdot k / (k - 2)$
Chi-squared	$\chi^2(k)$	Hypothesis testing, goodness of fit; non-negative, right-skewed	k	$2k$
Exponential	$Exp(\lambda)$	Rapid early change then plateau (video views)	$1/\lambda$	$1/\lambda^2$
Logistic	$Logistic(\mu, s)$	Continuous input $\rightarrow$ binary outcome (sports forecasting)	μ (location)	$s^2 \cdot \pi^2 / 3$

5.2 Key Formulas to Memorize

- **Universal variance relation:** $\sigma^2 = E(Y^2) - [E(Y)]^2$
- **Binomial probability:** $P(y) = C(n, y) \cdot p^y \cdot (1 - p)^{n-y}$
- **Poisson probability:** $P(y) = \lambda^y \cdot e^{-\lambda} / y!$
- **Standardizing:** $Z = (Y - \mu) / \sigma$
- **Discrete peculiarity:** $P(Y \leq y) = P(Y < y + 1)$
- **Continuous peculiarity:** $P(Y > x) = P(Y \geq x)$, because $P(Y = x) = 0$
- **68 - 95 - 99.7 law:** percentages of normal data within 1, 2, and 3 standard deviations of the mean

5.3 Review Questions

1. What is the difference between uppercase Y and lowercase y in probability notation?
2. Why is standard deviation often preferred over variance when interpreting spread?
3. A department survey covers every employee in that department. Is this population or sample data? Explain why it could be both.
4. Why do the mean and variance of a uniform distribution have no predictive power?
5. An unfair coin gets heads 70% of the time. If heads = 1, what are the expected value and variance of a single flip?
6. You guess all 10 questions on a true/false pop quiz. Which distribution models your total number of correct answers, and with what parameters?
7. A café receives 5 customer complaints per week on average. Which distribution and which formula would you use to find the probability of exactly 2 complaints next week?
8. Why is $P(\text{finishing a race in under 6 minutes})$ equal to $P(\text{finishing in at most 6 minutes})$ for continuous data?
9. How do you obtain the CDF from a PDF, and the PDF from a CDF?
10. A data set is normally distributed with $\mu = 100$ and $\sigma = 10$. Using the 68 - 95 - 99.7 law, what percentage of values fall between 80 and 120?
11. Convert the value $Y = 130$ from question 10's distribution into a Z-score. What does that Z-score mean in words?
12. You have only 12 observations of what should be normally distributed data. Which distribution should you use instead, and how does its shape differ?
13. For a chi-squared distribution with 6 degrees of freedom, what are the expected value and variance?

14. Daily downloads of a new app start high and decline rapidly. Which distribution fits, and what transformation makes this data suitable for linear regression?

15. In the FIFA 19 case study, the shot power histogram had two peaks. What caused this, and what was the fix?

16. In the membership example, what does the "location" of the logistic distribution represent, and what business decision did it support?

5.4 Answer Key

- 1.** Uppercase Y is the random variable — the actual outcome of the event. Lowercase y is one specific possible outcome.
- 2.** Variance is in squared units (e.g., seconds²), which has no direct interpretation. Standard deviation is in the same units as the mean, so it can be interpreted directly.
- 3.** It is population data for the department (all of its members) but only sample data for the whole company (a part of all employees). Population vs sample depends on what "all the data" means for the question being asked.
- 4.** Every outcome has the same probability, so the average tells you nothing about which outcome to expect — no value is more likely than another.
- 5.** $E(X) = p = 0.7$. Variance $= p(1 - p) = 0.7 \times 0.3 = 0.21$.
- 6.** Binomial: $X \sim B(10, 0.5)$ — 10 trials, 0.5 probability of guessing each question correctly.
- 7.** Poisson with $\lambda = 5$: $P(2) = 5^2 \cdot e^{-5} / 2! = 25 \times 0.0067 / 2 \approx 0.084$, roughly an 8.4% chance.
- 8.** Because the probability of any single exact value in a continuous distribution is zero — finishing in exactly 6:00.000... has probability zero, so including or excluding it changes nothing.
- 9.** CDF = the integral of the PDF from $-\infty$ to y . PDF = the first derivative of the CDF with respect to y .
- 10.** 80 to 120 is $\mu \pm 2\sigma$, so 95% of values fall in that range.
- 11.** $Z = (130 - 100) / 10 = 3$. The value 130 sits exactly 3 standard deviations above the mean — a rare outcome under the 99.7 part of the law.
- 12.** Student's T . Same bell shape and symmetry, but fatter tails to accommodate extreme values, which represent a larger share of a small sample.
- 13.** $E(Y) = k = 6$. Variance $= 2k = 12$.
- 14.** Exponential distribution. Take the natural logarithm of every element — the transformed data is much closer to normal and suitable for linear regression.
- 15.** Two distinct groups were mixed in one data set: outfield players and goalkeepers, whose key stats differ completely. The fix was splitting the groups — each cluster then behaved normally on its own.
- 16.** The location (≈ 34) is the age around which the probability of holding a premium membership rises sharply — interpreted as the age of financial stability. It supported the decision to target customers over 34 with premium offerings like legend cards.